

Ng Guoyou

ng.guoyou[at]gmail.com
github.com/gyng

Experience

SPH Media

SPH Media is the largest media organisation in Singapore.

Head, Digital Platform, Singapore

April 2024–Present

- Leading around 20 developers focused on developer tooling, platform services, content archival, and B2B content distribution. Together, these teams serve >50 software developers and have >\$1M SGD yearly revenue.
- Developed mid- and long-term strategies, roadmaps, and mandates for teams. Created engineering strategies to reduce cost, increase capabilities, and scale up the number of customer engagements with small development teams.
- Developed and implemented engineering policies and processes for teams around technical design, development, and operations.
- Set and maintained engineering standards and values around scalable + reliable architecture, automated testing, baseline code quality, and operations across teams.
- Guide (and sometimes create) technical and architecture designs as part of the technical design process.
- Reorganised teams as part of organisation transformation into more efficient structures.
- Work with neighbouring teams and users to prioritise and unblock development.
- Implemented hiring processes and pipelines.

Lead Software Engineer, Singapore

2022–April 2024

TypeScript, Node, Python, AWS, React, Vite, Docker, OpenSearch

- Led around 10 developers across two teams to build platform services.
- Architected, developed, and ran an in-house web deployment platform for SSR/ISR Vite/React applications designed for reliability on AWS (think Vercel). The platform features a boilerplate done using Vite, edge logic, edge middleware, incremental-static regeneration, a Backend-for-Frontend framework, and deployment workflows. This web platform serves millions of page views per day across >10 publications. Assisted teams with migration away from Drupal and WordPress setups.
- Architected, developed, and ran the content API and delivery service. This includes content indexing as a service using OpenSearch (think Algolia), ingestion pipelines, content delivery, and on-the-fly image optimisation services using sharp.
- Architected and developed the in-house browser tracking script, associated backend service, and session ID tracking scheme.

Basis AI (acquired by Aicadium)

Basis AI was a startup that built a machine learning platform for data science teams.

Software Engineer, Singapore

Jan 2019–Oct 2020, Jun 2021–2022

TypeScript, Python, Rust, Webpack, React, Redux, Docker, Kubernetes, UX

- Employee #3. Full stack developer on a machine learning platform for training and deployment of data science workloads.
- Main contributor to frontend with Webpack, TypeScript, React+Redux. Created reusable React component library.
- Built client workload monitor in Rust; includes streaming ZIP and large file uploads for workload artifacts.
- Developed backend features in Python such as pagination and RBAC.
- Implemented Docker image caching for fast workloads using kaniko.
- Built company website in Next.js.
- Conducted basic UX research and user testing.

Credit Suisse

Software Engineer, Singapore

Nov 2020–Jun 2021

TypeScript, React, Redux, MobX, Webpack

- Maintained and implemented features for internal financial frontends around securities order placement, disclosures, and compliance.

GovTech (Government Technology Agency)

GovTech is a Singapore government agency responsible for citizen-facing digital services.

Software Engineer, Singapore

Jan 2016–Jan 2019

TypeScript, React, Redux, Webpack, Python, Rust, Ruby on Rails

2018, [MyCareersFuture.sg](#), job portal

- Implemented frontend features for MyCareersFuture.
- Orchestrated codebase transition to TypeScript.
- Bootstrapped greenfield employer module with TypeScript.

2018, [OpenCerts.io](#), academic qualification verifier

- Created frontend tools to bulk-register academic transcripts on Ethereum using TypeScript and React.

2017, Data ingestion validator, API endpoint, and data visualisation application

- Reduced data ingestion time by 50 times.
- Developed frontend application for geographic data from scratch with IE11 support.

2016, [Business Grants Portal](#), grant application and tracking portal

- Implemented features for and fixed bugs on forms with complex validation logic.
- Integrated with SAML.
- Introduced linting for Ruby and JavaScript on a large codebase.

Roomorama

Intern, Software Engineer, Singapore

Summer 2013

Ruby on Rails

Education

Bachelor of Computing (Communications & Media), Hons.

National University of Singapore, 2011–2015

UX, HCI, and computer graphics focus

Personal

Singapore citizen. Served 2 years of National Service in the Singapore Armed Forces. Native English and Mandarin Chinese speaker.

Personal Projects

- [A zero-config, static, file-based album generator, 2022](#)
- [A now-playing desktop widget for Windows using Rust, Tauri, and Svelte, 2023](#)
- [TypeScript library to fetch file listings and individual files from a remote ZIP file, 2021](#)
- [Browser extension for organising downloads with >3000 users, 2018](#)
- [Retro image filtering application in React and Redux, 2017](#)
- [Chatbot for IRC, Slack, Discord in Ruby with a plugin system, 2016](#)
- [Toy audio synthesizer library in Rust with MIDI support and WebAssembly interface, 2014](#)
- [Sharing sessions and presentations](#)

More projects on [GitHub](#).